

# How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound

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## Abstract

Reduced (modal) deformable bodies resting in unilateral contact with rigid bodies are usually resolved by a velocity-level complementarity law that couples rigid and modal unknowns through one shared multiplier. Interactive solvers run that law at small, fixed local iteration budgets. We measure what this costs. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene  $\times$  relaxation  $\times$  budget sweep, we find sharply formulation-dependent behaviour. Against the measured gross rigid-side loss (scene-wide, gravity-corrected, and not a per-contact dissipation), the position-based host overdraws by up to  $4.4 \times 10^7$  J —  $1.2 \times 10^5$  times the incident rigid kinetic energy in the most starved cells — the augmented-Lagrangian host overdraws in 23 of 24 cells but by at most 15 J, and the implicit realization never overdraws at all. Pervasive but negligible and rare but catastrophic are different pathologies, and the customary ratio-to-incident-energy diagnostic does not separate them. A budget sweep shows the position-based amplification is a truncation artifact: it decays monotonically toward a converged reference computed in the same code path, and the gap to that reference shrinks by six orders of magnitude. Because the converged regime is unaffordable at interactive budgets, we add a cumulative modal-storage bound funded by that measured gross rigid-side loss, enforced by a reservoir ledger and a projection of the realized modal state, which holds in all 90 measured cells (78 distinct configurations). We also measure what that projection costs contact validity, and find worst-case post-projection penetration of 21.6 mm — 72% of the supporting board's thickness.

## CCS Concepts

• Computing methodologies → Physical simulation.

## Keywords

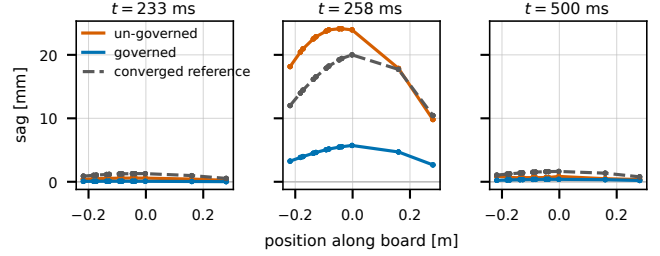
contact, modal reduction, passivity, position-based dynamics

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## 1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with



**Figure 1: The deflected support surface through one impact at a starved budget (8 $\times$ 2), sampled at the 48 support contact rows:  $d_i = U_y[i]^T \mathbf{q}$ , the quantity (1) sees. Un-governed, the board reaches 24.1 mm of sag against the converged reference's 20.0; governed, the bound holds peak modal energy to 29.3 J but takes the legitimate sag down with the spurious ring, to 5.9 mm. A non-diverging result, not a faithful one (§3.3).**

rigid bodies, the established treatment resolves rigid and modal unknowns together through a shared contact multiplier at the velocity level [12]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [1, 6]. Interactive modal deformation subject to collision and contact constraints predates both [5].

We take that coupled row as given. Written at position level it is one linearization step from the velocity-level law: for gap function  $C$ ,  $C^{n+1} \approx C^n + h \mathbf{J} \mathbf{u}^{n+1}$ , which divided by  $h$  is the velocity-level complementarity condition with a  $C^n/h$  stabilization bias, carrying the same Jacobian and the same multiplier as Zheng and James [12]. *The row is not our contribution*, and ours is narrower than theirs: modes couple through normal rows only, and the position-level form takes  $e = 0$  restitution.

Interactive solvers do not solve that row to convergence. They run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [10]. *That observation is not ours either*. Velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [7]. What is missing is a quantitative answer for this specific case — unilateral contact-to-modal transfer — and a mitigation that fits inside an interactive budget.

### Contributions.

- (1) **Measurement.** One contact row, three fixed-budget solver formulations, one identical 24-cell adversarial sweep, one machine (§3.1); plus an iteration-budget sweep against a converged reference showing the position-based injection is a truncation artifact (§3.2).

- (2) **A cumulative storage bound.** A one-directional bound on energy entering modal storage, funded by measured gross rigid-side loss and enforced by a reservoir ledger and a projection of the realized modal state (§2); it satisfies the prescribed inequality in all 90 measured cells (78 distinct configurations).
- (3) **The cost of enforcement.** Post-projection contact validity — penetration, corrective impulse, multiplier variance, momentum — measured rather than assumed (§3.3).

*Relation to concurrent energy-control work.* Wei et al. [10] guarantee discrete passivity for any finite inner-iteration budget by construction, via Douglas–Rachford splitting in wave coordinates whose Fejér monotonicity supplies algorithmic dissipation; their interfaces are *bilateral* couplings, not unilateral contact, and the guarantee is structural rather than a budgeted transfer cap. You et al. [11] modify implicit Euler to track user-specified *total-energy* targets bidirectionally on full nonlinear elastodynamics with barrier contact. Ours is neither: a one-directional cap on what contact may deposit into modal storage, funded by measured gross rigid-side loss, enforced by scaling realized modal state. For flexible bodies in a game engine, Zobel et al. [13] solve ray-traced penalty contact quasi-statically without a shared unilateral multiplier; modes and modal derivatives have been used in a position-based host as a *deformation* surrogate [8], whereas here the modal amplitudes enter as contact DOFs driven by the shared multiplier.

*Relation to passivity control and energy projection.* The governor transplants an established control idea rather than inventing one. Time-domain passivity control [4] damps once an energy observer goes negative; energy-tank formulations [3] give that observer explicit state, so a controller spends only what its tank holds; Rath [9] enforces exact per-interface energy stability for contacting modal objects at audio rate, where we cap a scene-level scalar reservoir. We keep that structure and change two things: the tank is funded by *measured gross rigid-side loss* — energy the solve removed elsewhere — rather than by the port it regulates, and the actuator scales realized *state* rather than modulating a force, because a fixed-budget solver commits its multipliers before the excess is observable. FEPR [2] also projects state post-solve, but onto conservation of *total* energy; ours is one-sided and never restores energy the solver lost.

## 2 The row and the cumulative bound

Let  $\mathbf{q} \in \mathbb{R}^r$  be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is  $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$ . A support-contact row between a rigid corner at world height  $y_c$  and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with  $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$ , so one multiplier  $\lambda$  drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law.

*The invariant.* We require that the energy stored in the modal subsystem never exceed the *measured gross rigid-side loss* — defined exactly at (3) below — cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1], \quad (2)$$

The transfer efficiency  $\eta$  sets how much of the measured loss may be spent; *every experiment in this paper runs*  $\eta = 1$ , the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total stays inside the budget, which is what lets a legitimate ring persist across substeps.

*Every term, exactly.* Modal storage is the mechanical energy of the reduced state as written above, and  $E_{\text{mod}}^0 = 0$ : every scene starts from an undeformed, resting support, so the run’s own settling transient must be funded like anything else. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g}(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where  $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \boldsymbol{\omega}_b^\top \mathbf{I}_b \boldsymbol{\omega}_b]$  is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and  $\pm$  denote the two ends of substep  $k$ . We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid–rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§4). Gravity enters only through the work term  $W_g$ , which makes free ballistic motion register zero supply: a falling body’s kinetic gain exactly equals the gravity work, so nothing is credited and only an inelastic velocity change funds the reservoir. Endpoints are captured at substep boundaries and tile the timeline exactly, so no rigid energy change escapes the accounting.

*What is guaranteed, and what is not.* Equation (2) is a *cumulative, gross-loss-funded storage ceiling on the modal subsystem*, with a supply that is *measured, not prescribed*: the right-hand side is the measured gross rigid-side loss of (3), never a tuned constant or a fraction of incident energy. It is not contact-port passivity and not a signed per-interface transfer bound — it constrains a scalar reservoir, not a port. The *gross* sum ignores substeps in which the rigid subsystem *gains* energy, so those cannot cancel earlier credits, and energy returned and re-dissipated counts as fresh supply; §4 measures both.

*The enforced inequality is slightly weaker than the stated one.* Reported violations use (2) exactly as printed. The implementation’s own test additionally forgives one substep’s largest deposit,  $\max_k \eta \max(\Delta E_{\text{rig}}^k, 0)$ , absorbing the same-substep granularity artifact of the loop above: modal potential energy can rise in the very substep whose credit funds it. In these scenes that allowance is worth 7–389 J — immaterial where the overdraw is  $10^5$  J or more, decisive where it is not: under it the augmented-Lagrangian host would be recorded as violating 1 of 24 cells rather than 23. We report the stricter reading throughout, for both governed and un-governed columns, so that one inequality adjudicates both; the governed runs satisfy it with a worst margin of  $1.1 \times 10^{-13}$  J.

*Enforcement.* A reservoir  $B$  accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, in the same substep:  $\Delta E_{\text{rig}}$  is not observable until the velocity solve has finalized the rigid velocities, so the substep credits  $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$  at that point, tests the realized modal energy against it, and only then debits  $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$ . Crediting within the

funding substep is precisely what lets modal potential energy rise in the substep that pays for it — the granularity artifact the forgiveness above absorbs. If the realized modal energy would overdraw  $B$ , the realized state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma (\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma = \min\left(1, \sqrt{\frac{E_{\text{mod}}^- + B}{E_{\text{mod}}^+}}\right), \quad (4)$$

One substep of the governed loop, identical in all three hosts:

- (1) snapshot  $E_{\text{rig}}^-$  (pure KE, all bodies),  $E_{\text{mod}}^-$ ,  $\mathbf{x}^-$ ;
- (2) predict (gravity); generate contacts at the predicted pose;
- (3) position solve ( $K$  iterations), then velocity solve — *the contact solve*;
- (4)  $\Delta E_{\text{rig}} = (E_{\text{rig}}^- - E_{\text{rig}}^+) + W_g$ , i.e. (3), post-solve;
- (5) credit  $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$  — same substep;
- (6) if  $E_{\text{mod}}^+ > E_{\text{mod}}^- + B$ , scale the state by  $\gamma$  of (4);
- (7) debit  $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$ . Contact is **not** re-solved.

with  $E_{\text{mod}}^\mp$  the modal energy before and after the solve. The form is closed because  $E_{\text{mod}}$  is quadratic in the state: scaling  $\mathbf{q}$  and  $\dot{\mathbf{q}}$  together scales both the kinetic and the elastic term by  $\gamma^2$ , so  $\gamma$  bounds the total rather than only the velocity. Two consequences matter and are measured below: the projection is inert whenever the solve is already within budget ( $\gamma = 1$ , bitwise-identical trajectories), and it does not re-solve contact (step 7).

### 3 Results

Every solver-behaviour number below was produced on one machine (Table 1); the sole exception is the device-resident timing paragraph of §3.5 (NVIDIA RTX 4090), reported separately. All are frozen with generating commands and commit hashes in the supplemental ledger. Cross-machine mixing is deliberately avoided: chaotic contact stacks diverge between architectures under floating-point reassociation.

#### 3.1 One row, three formulations, the same 24 cells

We drive an impactor into a modally-reduced support in three scenes — a shelf board, a ledge slab, and a  $2.2 \times 1.1$  m table, the last two following the layouts of Coevoet et al. [1] — and sweep modal relaxation and the local budget (iterations  $\times$  substeps) over  $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$ . Table 1 lists what the hosts do differently and what is pinned identical. Two entries deserve emphasis. The relaxation axis is *inert* on the impulse host — its implicit modal weight needs no under-relaxation and the attribute is never read — so its two relaxation rows are bit-identical by construction, not coincidence. And warm start differs. We run each host as published:  $\lambda \leftarrow 0$  every substep is the position-based host’s own form, and carried duals with a growing penalty are constitutive of the augmented-Lagrangian method, so neither is a knob we chose; the impulse host’s per-row  $\lambda$  cache is the one genuine policy default among the three. The comparison is therefore as-deployed, not compliance-matched.

*Equal row evaluations, unequal outcome.* A cell costs  $K \cdot S$  row evaluations per frame, so the budget axis spans  $\times 64$  — steeper than its labelling suggests — and conflates two refinements that are not interchangeable. At a fixed 32 row evaluations on the shelf drop, spent as iterations ( $K=32$ ,  $S=1$ ) the ratio is 0.300 and (2) holds; spent as substeps ( $K=4$ ,  $S=8$ ) it is 3.13, overdrawing by 481 J: substep

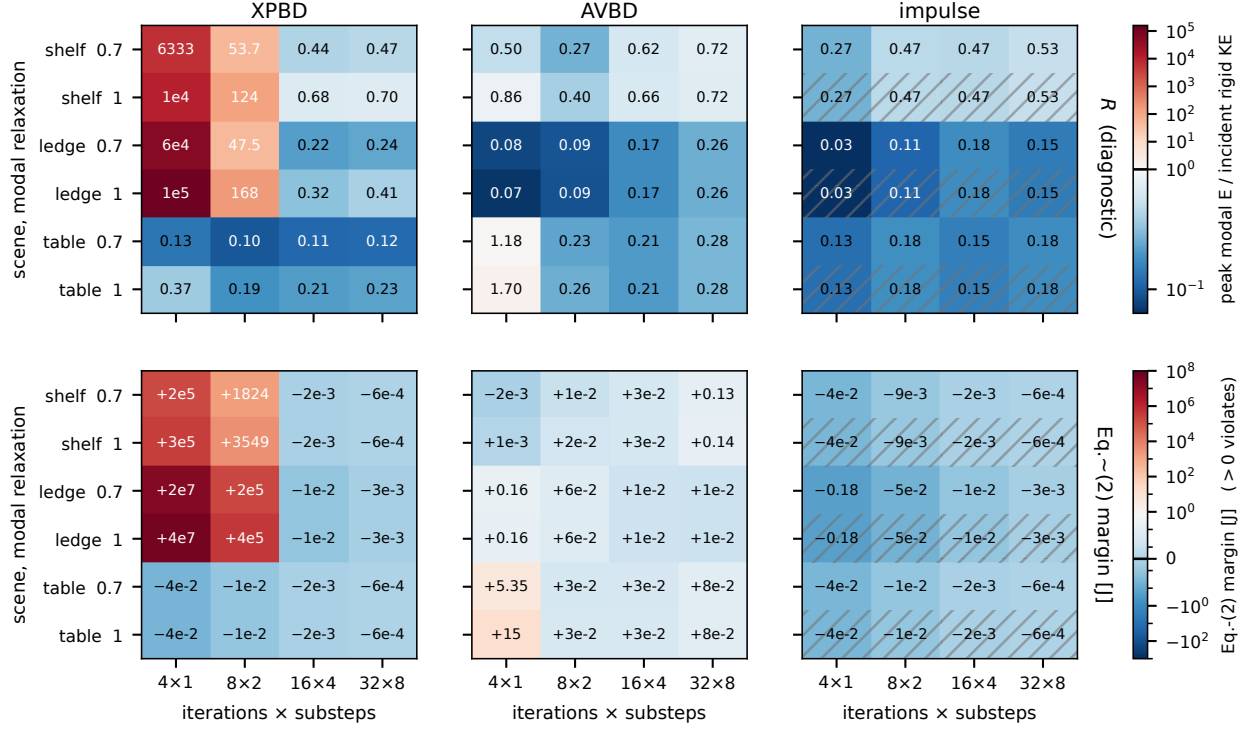
**Table 1: What differs between the three hosts, and what does not. Everything below the rule is held identical; everything above it is intrinsic to the formulation. Entries are read from the implementation, with source anchors in the supplemental ledger.**

|                    | position-based                                                                                                     | augmented-Lagr.                                     | seq. impulse                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unknown solved for | position                                                                                                           | position + multiplier                               | velocity                                                                                                                                                                |
| contact treatment  | compliant projection, $\tilde{\alpha} = \alpha/h^2$ ( $\alpha=0$ : rigid)                                          | penalty + dual update, $\alpha=0.99$ , $\beta=10^5$ | Schur sys-tem, $\mathbf{A} = \frac{c_{\text{fm}}}{h^2} \mathbf{I} + \mathbf{J} \mathbf{M}^{-1} \mathbf{J}^\top + \tilde{\mathbf{G}} \mathbf{W} \tilde{\mathbf{G}}^\top$ |
| modal weight       | explicit; per-mode compliance $1/(H_{ii}h^2 - 1)$                                                                  | block-GS, full Newton step on $\mathbf{q}$          | <i>implicit</i> , $\mathbf{W} = (\mathbf{M} + h\mathbf{D} + h^2\mathbf{K})^{-1}$                                                                                        |
| iteration          | GS sweeps, gap re-evaluated each sweep                                                                             | coloured primal GS + $\mathbf{q}$ -block            | PGS over rows                                                                                                                                                           |
| warm start         | none ( $\lambda \leftarrow 0$ each substep)                                                                        | $\lambda$ and penalty carried over                  | $\lambda$ cache keyed per row                                                                                                                                           |
| relaxation         | 0.7 on $\mathbf{q}$ and the support block                                                                          | 0.7 on the $\mathbf{q}$ -block Newton step          | <i>inert</i> : never read                                                                                                                                               |
| $h$ ; substep      | 1/120 s; $h/S$ , contacts regenerated per substep                                                                  |                                                     |                                                                                                                                                                         |
| modal rank $r$     | 24 (shelf), 28 (ledge), 24 (table)                                                                                 |                                                     |                                                                                                                                                                         |
| modal damping      | Rayleigh $\mathbf{D} = \alpha_0 \mathbf{M} + \alpha_1 \mathbf{K}$ , $\alpha_0 \in \{2, 3\}$ , $\alpha_1 = 10^{-5}$ |                                                     |                                                                                                                                                                         |
| stepper; $\eta$    | implicit midpoint; $\eta = 1$                                                                                      |                                                     |                                                                                                                                                                         |
| machine            | Apple M4, CPU only, CPython 3.12                                                                                   |                                                     |                                                                                                                                                                         |

refinement alone does not converge the row (full ladder in the supplement).

Figure 2 shows the outcome. The position-based host exceeds  $R = 1$  in 8 of 24 cells, worst case  $1.19534 \times 10^5$  (ledge, relaxation 1.0,  $4 \times 1$ : peak modal energy  $4.44 \times 10^7$  J against an incident 371 J). The augmented-Lagrangian host exceeds  $R = 1$  in 2 of 24 — both at the starved  $4 \times 1$  corner of the table scene, worst 1.70 — so it is *not* unconditionally benign either. The ordering is scene-dependent and it inverts: worst-over-scenes at  $4 \times 1$  the position-based host is five orders of magnitude more energetic ( $1.2 \times 10^5$  vs 1.70), but *on the table scene itself* the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

*The ratio is not the invariant.* One impactor’s incident kinetic energy is not the supply (2) budgets against, so we measure (2) directly, un-governed, on all three hosts, running the reservoir accounting live while forcing  $\gamma \equiv 1$ : every state write in the enforcement path is guarded by  $\gamma < 1$ , so the trajectory is bit-identical to an un-governed run (the ratios above reproduce the independent sweep exactly). We report the signed margin of (2) in joules; positive violates. The two metrics agree on the position-based host (8/24 either way) and on the implicit one (0/24, margin at worst  $-5.7 \times 10^{-4}$  J), but they part company on the augmented-Lagrangian host, which violates (2) in 23 of 24 cells while  $R$  flags only 2. The magnitudes separate the two failures: in 21 of its 23 cells the augmented-Lagrangian overdraw is under 0.15 J (worst 15.1 J), against the position-based host’s  $4.4 \times 10^7$  J. Pervasive but negligible overdraw and rare but catastrophic overdraw are different pathologies, and only the invariant distinguishes them.



**Figure 2: The same contact row in three fixed-budget solver formulations over an identical 24-cell sweep (3 scenes × modal relaxation {0.7, 1.0} × budget), governor OFF. Top: the ratio  $R$  = peak modal energy / peak incident rigid kinetic energy, log scale diverging at 1 – a severity diagnostic. XPBD exceeds 1 in 8/24 cells, AVBD in 2/24, the implicit realization in 0/24. Bottom: the invariant itself – the signed margin of (2) in joules, symmetric-log, diverging at 0; positive violates. The two rows disagree on AVBD. Hatched rows mark the relaxation axis, which that formulation does not consume; those rows are bit-identical by construction and shown for grid symmetry, not as independent measurements. We report joules rather than a second ratio because every candidate denominator misleads: supply-so-far explodes in the opening substeps, the run total hides early peaks.**

With enforcement enabled, all 72 cells of this sweep satisfy (2) – 60 of them distinct configurations, since the implicit host does not consume the relaxation axis and its 12 relaxation pairs are bit-identical by construction (Fig. 2); the 18 deployed-budget cells of §3.2 satisfy it too. The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell. At the well-resolved budgets (16×4, 32×8) the projection fires zero times on the position-based host at relaxation 0.7: it is inert exactly where the solve is already within budget.

### 3.2 The position-based amplification is a truncation artifact

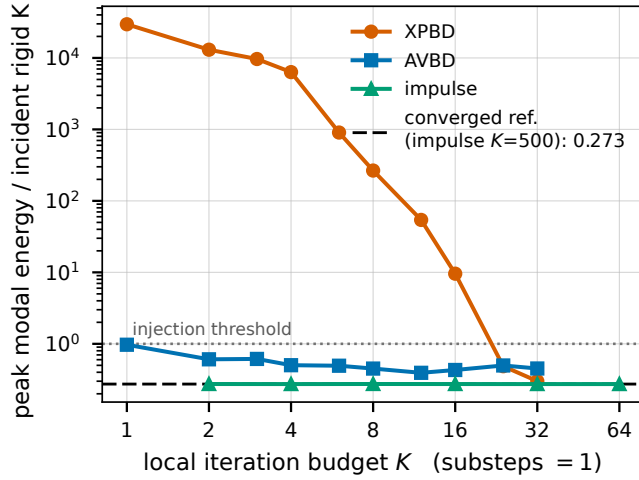
Figure 3 isolates the local iteration budget. The position-based host falls monotonically from  $2.96 \times 10^4$  at  $K=1$  to 0.300 at  $K=32$ , approaching the converged reference 0.2735; the gap  $|\text{XPBD} - \text{reference}|$  shrinks from  $2.96 \times 10^4$  to  $2.7 \times 10^{-2}$ . The implicit realization moves only in the fourth decimal across  $K = 2 \dots 500$  (spread  $4.9 \times 10^{-4}$ ).

That the energy ratio falls with  $K$  does not by itself show the row is converging – it could be one scalar coincidentally shrinking. Measuring the constraint directly settles it: the end-of-substep complementarity residual  $\|\min(C, \lambda)\|_\infty$  on the same sweep falls

monotonically from  $2.79 \times 10^{-2}$  at  $K=1$  to  $3.56 \times 10^{-6}$  by  $K=64$ , where it bottoms out ( $K=128$  moves it in the seventh significant figure). The energy crossing at  $K \approx 24$  coincides with the residual passing  $\sim 3 \times 10^{-5}$ ; the amplification disappears exactly as the complementarity conditions start to hold. We report this for the position-based host only: the augmented-Lagrangian multiplier is not the same object, and the implicit realization is already converged at  $K=2$ , so neither yields a comparable curve.

*Self-convergence stops short of the reference.* Pushing the position-based host to  $K=500$  against itself does not close the remaining gap: its ratio plateaus by  $K \approx 64$  (0.3003, 0.2999, 0.2997, 0.2996, 0.2996 at  $K = 32 \dots 500$ ), so the residual  $2.6 \times 10^{-2}$  against the oracle’s 0.2735 is a fixed point, not a tail – the gap improves by only 1.03× over that range. The state agrees no better: on the deflection trajectory  $d_t(t) = U_y[i]^T q(t)$  the peak matches to 2.4% but the trajectories differ by  $\|d^{500} - d^{\text{oracle}}\|_\infty = 6.8$  mm, 34% of the oracle’s own peak, likewise flat from  $K=32$ . The two hosts converge to different discrete solutions – expected, since they discretize the same continuous law differently – so the sweep separates two effects easily conflated: a truncation pathology convergence removes, atop a formulation difference it does not. (No ring frequency is reported here: these





**Figure 3: Energy ratio versus local iteration budget  $K$  on a deterministic shelf drop, substeps pinned at 1 so  $K$  is the only variable, governor OFF. The position-based host is monotone non-increasing in  $K$  and decays toward the converged reference, crossing the incident-energy threshold between  $K=16$  and  $K=24$ ; the gap to that reference shrinks by six orders of magnitude. The reference is the implicit realization at  $K=500$  — the same code path run to convergence, not a different model. That realization is already converged at  $K=2$ . The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small  $K$ , so it must not be read as a convergence trend.**

windows are dominated by quasi-static sag, so the dominant spectral peak moves with window length; the resolved comparison is in §3.4.)

This is the central diagnostic result, and it cuts both ways. It confirms that the amplification is created by truncation and largely cured by convergence — but it equally says that iterating harder, or adopting the implicit formulation, would remove the pathology outright. The bound of §2 is therefore not a substitute for convergence; it is for the regime where neither escape is available: interactive position-based solvers ship budgets near  $1 \times 8$  to  $2 \times 4$ , well below the  $K \geq 24$  this scene needs. Measured directly on 18 further cells (3 scenes  $\times$  3 hosts  $\times$   $\{1 \times 8, 2 \times 4\}$ , relaxation 0.7), the position-based host violates (2) in *all six* of its deployed cells, worst  $R = 2282$  — on 8 row evaluations, twice the  $4 \times 1$  corner’s 4. The deployed half of the budget trade is the worse half. The other hosts behave as in the sweep (5/6 and 0/6 cells, the former by at most 0.16 J). Governed, all 18 satisfy (2) (worst  $R = 1.022$ ), and the projection is gentler than at  $4 \times 1$ : worst penetration 9.8 mm against 21.6. Our implicit backend is a reference implementation whose cost we do not claim is representative of its formulation class.

### 3.3 What the projection costs

**Accuracy.** Bounding the energy is only useful if the bounded answer beats the unbounded one. At a moderate injecting cell (shelf,  $8 \times 2$ , relaxation 0.7), peak modal energy is 1555.6 J ungoverned

**Table 2: Post-projection contact validity at starved budgets, per clamp-active substep. Gap violation is measured immediately after the  $\gamma$ -scale; the corrective impulse is the next substep’s normal impulse against the unclamped steady-state median, reported as a ratio because the hosts’ multipliers are different objects. The  $4 \times 1$  position-based cells clamp on nearly every substep, so no steady state exists there (—). The augmented-Lagrangian rows are its only two cells with  $R > 1$ .**

| scene                                 | budget       | clamped | gap viol. med / max [mm] | impulse       | $\lambda$ var. |
|---------------------------------------|--------------|---------|--------------------------|---------------|----------------|
| <i>position-based, relaxation 0.7</i> |              |         |                          |               |                |
| shelf                                 | $4 \times 1$ | 107/108 | 1.25 / <b>21.6</b>       | —             | —              |
| shelf                                 | $8 \times 2$ | 162/216 | 0.30 / 6.46              | $6.90 \times$ | $26 \times$    |
| ledge                                 | $4 \times 1$ | 98/108  | 0.07 / <b>21.2</b>       | $1.03 \times$ | $1.07 \times$  |
| ledge                                 | $8 \times 2$ | 96/216  | 0.07 / 4.08              | $8.69 \times$ | $58 \times$    |
| <i>augmented-Lagrangian</i>           |              |         |                          |               |                |
| table (0.7)                           | $4 \times 1$ | 27/108  | 0.01 / 1.43              | $1.17 \times$ | $0.41 \times$  |
| table (1.0)                           | $4 \times 1$ | 25/108  | 0.02 / 3.11              | $1.39 \times$ | $1.38 \times$  |

and 29.26 J governed against the converged reference’s 7.92 J: the energy error falls  $196 \times \rightarrow 3.7 \times$ , or  $+1548 \rightarrow +21$  J. Against the host’s own fixed point (8.22 J) it is  $189 \times \rightarrow 3.6 \times$ : the result does not depend on the reference chosen. The trajectory, however, gets worse:  $\|d - d^{\text{ref}}\|_{\infty}$  is 6.5 mm ungoverned and 14.3 mm governed, 33% and 71% of the reference’s peak: a safety envelope, not an accuracy device. At the deployed  $1 \times 8$  cell the same comparison gives  $2823 \times \rightarrow 3.7 \times$  on energy and the trajectory improves as well ( $108 \rightarrow 96\%$  of reference peak) — the trade is budget-dependent, but 96% is still a wrong trajectory.

The reason also explains how a  $196 \times$  energy error coexists with a peak deflection only  $1.2 \times$  too large. The excess is not kinetic (0.3%) and not low-frequency: this scene’s spectrum is bimodal (ten bending modes below 2.1 kHz, six stiff ones above 20 kHz), and 99.6% of the ungoverned energy sits in the stiff cluster,  $\sim 200 \times$  above the 240 Hz substep rate, against 0.0% for both references. Being one scalar,  $\gamma$  cannot redistribute across the spectrum: it fixes the *amount* ( $1555.6 \rightarrow 29.3$  J), leaves the character intact ( $99.6 \rightarrow 97.7\%$ ), and scales the legitimate sag down with the noise ( $24.1 \rightarrow 5.9$  mm against 20.0).

**Contact validity.** Since  $\gamma < 1$  shrinks the sag  $U_y^T q$ , (4) lifts the support *into* the resting body without re-solving contact. Table 2 reports the consequence, instrumented by runtime wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.25 mm, but the worst case reaches 21.6 mm, a  $3.5\text{--}12.6 \times$  increase over its pre-scale value. The shelf board is 30 mm thick, so that is 72% of its thickness; against the unclamped scene at a converged budget (resting sag 1.7–2.4 mm, peak dynamic deflection 20.0–20.5 mm) it is  $9\text{--}13 \times$  the sag and  $1.05\text{--}1.08 \times$  the peak the board ever reaches. At its worst substep the projection removes essentially the entire deflection — a visible interpenetration, not a sub-millimetre artifact, and it occurs where the projection is most load-bearing. Contact recovers in the next substep rather than accumulating, but that correction costs a normal impulse up to  $8.7 \times$  the steady-state value, with multiplier variance up to  $58 \times$  higher. Net rigid linear momentum is unchanged across the projection to exactly zero in every clamp-active substep

of every cell, necessarily so since (4) writes only modal state; we verified it numerically rather than asserting it.

The same instrumentation on the augmented-Lagrangian host, at its only two cells with  $R > 1$ , finds the projection an order of magnitude gentler: a quarter of substeps rather than nearly all,  $\gamma$  bottoming at 0.60, worst-case violations of 1.4 and 3.1 mm against 21.6, and corrective impulse and multiplier variance within  $1.4\times$  of steady state where the position-based host reaches  $8.7\times$  and  $58\times$ . This is what §3.1 predicts — its overdrafts are at most 15 J — while the *relative* effect is comparable ( $3.1$  and  $5.5\times$  against  $3.5$ – $12.6\times$ ). The hosts differ in the scale of the correction, not its character.

### 3.4 The reduced response against a full-FEM reference

Bounding the energy is only useful if the response being bounded is the right one. On the ledge scene we compare the reduced arm against an unreduced FEM solve of the same mesh, both built on the *same* discrete operator (the reduced basis is  $\text{eigsh}(\mathbf{K}, \mathbf{M})$  of the operator the reference integrates), so discretization bias cancels from every ratio. The reduced-to-reference peak-deflection ratio rises monotonically  $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$  as  $h$  refines  $1/120 \rightarrow 1/960$ , plateauing near 0.9; the residual  $\sim 10\%$  is  $k=24$  mode truncation on a thick, high-frequency slab, not a coupling error — the ring frequency already agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees too: peak deflection sits at the pedestal antinode rather than under the impact, and the falloff tracks the reference at Spearman  $\rho = 0.89$  with no fitted attenuation term. This is the pre-topple, small-displacement regime; we claim nothing beyond it.

### 3.5 Cost

On the CPU host at  $16\times 4$ , over ten repetitions of 100 frames, the baseline step costs 11.0–125.6 ms by scene and formulation and the ledger adds 0.23–0.41 ms —  $0.9$ – $3.4\%$  — on the four shelf and ledge configurations, where it resolves above run-to-run variance. On the table scene it does not: the overhead's per-repetition spread exceeds its mean in both hosts ( $+1.35 \pm 2.08$  and  $+0.04 \pm 1.43$  ms), so we report it bounded by  $\sim 1.4$  ms rather than quote a figure. The percentage is the portable quantity: on a second machine absolute times differ by  $3.5$ – $4.6\times$  while the percentages agree to within 0.02 points wherever both resolve. These baselines are  $1.3$ – $15\times$  short of a 120 Hz budget, though shelf and ledge fit one at  $2\times 4$  with the governor on.

A separate device-resident path (NVIDIA RTX 4090), which carries the ledger read-only as a monitor rather than enforcing it, reaches 5.0–8.9 ms/step at  $16\times 4$  across four scenes (the two above, a  $2.5 \times 1.5$  m road slab under a dropped crate, and the table). The shelf and ledge scenes meet a 120 Hz budget there with margin ( $1.7\times$  and  $1.3\times$ ); the road scene sits at the boundary ( $1.01\times$  on the mean, with a 9.3 ms worst step); the table scene does not ( $0.94\times$ ). All four meet it at  $16\times 2$  or below. We make no unqualified real-time claim. A device-resident *enforced*  $\gamma$  does not exist in this work, so governed device timings are outside what we measure rather than a result we are withholding (§4).

## 4 Limitations

**Stable is not accurate.** Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory *away* from the converged reference even as it moves the energy toward it, for the structural reason given in §3.3. The result is bounded and non-diverging, not faithful; the honest use of the bound is as a safety envelope for budgets that would otherwise diverge. The identified next mechanism is to scale deviations about the quasi-static equilibrium rather than the whole state, preserving the load-bearing sag; we have not built it, and its unscaled equilibrium term would leave (2) holding only while the reservoir exceeds the sag's own strain energy.

**Convergence is the better cure where affordable (§3.2).**

**Scope of the row.** Normal rows only — friction is body-local, so scraping and rolling transfer are outside this coupling — and the position-level form fixes  $e = 0$ .

**The bound governs modal storage only,** not energy created spuriously in rigid degrees of freedom. In the impulse realization we observe box–box rows rectifying a support ring into net rigid motion; the ledger is blind to this by design.

**The supply is a gross sum, so it can be recycled.** Because the reservoir is funded by  $\sum_k \max(\Delta E_{\text{rig}}^k, 0)$ , energy returning from the modes to the rigid bodies and dissipated there is credited twice. We bound the exposure by accumulating the opposite channel,  $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$ , as a fraction of the supply: 0.4–27% (implicit) and 3–32% (position-based) — not the  $\leq 1\%$  we expected, so the caveat rests on measurement. On the augmented-Lagrangian host it reaches 102–118%: the rigid subsystem's gross *gain* exceeds its gross loss in every cell, the same phenomenon as the box–box rectification above rather than a property of the bound. There the supply is an upper envelope on genuinely dissipated energy, and a signed or per-interface accounting — which this scalar reservoir deliberately is not — would be needed to tighten it.

**Enforcement is host-side** and single-machine (CPU float64). A device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work.

**Formulation coverage.** Three formulations, three scenes, one contact regime; the augmented-Lagrangian host's small but near-universal overdraft makes "descent-class solvers are passive here" an empirical statement about this sweep, not a theorem.

## 5 Conclusion

Transcribing an established rigid-modal contact law into fixed-budget solvers is not energy-safe, and how unsafe depends strongly on the formulation: worst-case modal-to-incident ratios are  $1.2 \times 10^5$ , 1.70 and 0.53 across the three hosts on the same 24 cells. The position-based amplification is a truncation artifact convergence removes; the augmented-Lagrangian overdraft we measure but do not trace. Where convergence is unaffordable, a cumulative storage bound funded by measured gross rigid-side loss holds in every measured cell, at a contact-accuracy cost we quantify rather than omit.

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